Lecture 2 ERS312 Notes

<u>Ocean Acidification</u>

- Pleistocene: time of the ice ages (past 2.5 million years)
- Holocene Epoch: present times
- Ocean pH remains stable throughout Pleistocene
- Start of industrialization, more acidic

<u>Marine Pollution</u>

- Garbage patches that form (accumulation of plastic and garbage in ocean *Pacific Ocean)

<u>Overtourism</u>

- Overcrowding at ocean areas

<u>Overfishing</u>

- more pressure on marine ecosystems as population increases
- Amount of fish removed is regulated
- Grand banks are shallow area of ocean
- Warm and cold current meeting creating productivity (nutrients, plankton)
- Cod fisheries collapsed in 1992 in NFL
- Rapid increase of fishing after 1950s (new technology was designed that can detect fish swarms)
- So many fish taken out that ecosystem wasn't stable enough (catch limits weren't strict enough resulting in collapse)
- Older fish=more eggs it can lay (female)
- In newfoundland many of the older fish were taken so it was harder to replenish

<u>Earth's Surface</u>

- 70.8% of earths surface is ocean and 29.2% is land
- Less than a century of detailed info on oceans

4 Oceans +

- Largest= Pacific Ocean (deepest) *most peaceful of all
- Atlantic Ocean
- Indian ocean
- Arctic ocean is smallest
- Southern ocean (characterized by oceanographers) *all ocean are that lies south *not an actual ocean basin

Relative Ocean Size

- Arctic is smallest

What are Sea's?

- Smaller and shallower than oceans
- Always salt water
- Usually are enclosed by land and directly connected to an ocean
- E.g. Labrador Sea, Hudson Bay, Beaufort Sea

Average Ocean depth

- Pacific close to 4000 m deep
- Atlantic 3844m deep
- Arctic is most shallow
  *remember 3700m for average depth of oceans
  *max depth is 11km

Mariana Trench-Deepest Part of the Ocean

- Challenger deep (is deepest part)
- Ocean tremch is due to plate tectonics

Subduction Zones

- Oceanic plates colliding and results in one plate going down (pacific plate is old and cooled down; initially hot when young so old plate goes underneath young plate)

- place where they meet is a depression

Bathyscape Trieste

- The first to reach the challenger deep
- Lead pellets are released, and submersible becomes light again
- Gasoline lighter than water (air filles also lighter) *pressure is the problem
- Fill it with liquid that's lighter than water so it can rise again once you get rid of ballast

Canadas share of the oceans

- By far largest coastline (200km)
- Canada has access to 3 oceans (Atlantic, pacific, arctic ocean)
- ocean usage is very important
- faces big problem with respect to ocean passage rights (northwest passage *sea way up north)
- some other consider northwest passage as international waters

Who has the right to pass NW passage?

Strait

- narrow waterway which connects two larger bodies of water

Transit Passage

- allows freedom of navigation between one part of high seas to another
- both countries must agree to let ships through

Internal Waters

- NW passage apart of internal waters
- extension on land territory
- country can decide whether they want a ship to go or not

Why important?

- it’s a shortcut
- if a ship wants to go from Netherlands to Japan (can take NW passage)
- Interest by mining industries (cannot get there easily by ship which is the issue)

Why does Canada not want free passage through NW passage?

- Arctic waters are not charted to modern standards (no proper charted map) *risk for accident
- If accident occurs, rescue capabilities are poor
- No ports which leads to environmental concerns such as oil spills from oil tankers
- Increased soot from heavy fuel oil (dirty diesel oil) *soot will make surface of ice darker; melting of ice will be stronger

How to navigate the oceans?

Geographic coordinate system

- For navigating the ocean
- Divided by meridians and parallels by the equator
- Meridians are longitude
- Prime meridian (0 longitude; located in Greenwich England)

Why is determining lat/long so important on the ocean?

- A way to orient yourself
- Earths center, straight line to equator then your location= latitude
- Earths center, prime meridian to location= longitude

How did captains determine latitude?

- Angle along horizon and direction of North star
  *in southern hemisphere you use southern cross for reference

How to use a sextant?

- Use telescope to look along horizon and use mirror until you see sun in mirror reflecting on to another lens

How did captains determine latitude?

- No shadow cast means the sun is directly above you (trophic of cancer)

Tilt of Earth Axis

- earth axis is tilted by around 23 degrees

Reference lines on Earth

- March/September Equinox (sun hits earth directly on earth perpendicular)
- June equinox (sun rays hit perpendicular to tropic of cancer, tilt of earth axis)
- December solstice (sun hits earth perpendicular at trophic of Capricorn)

How did captains determine longitude?

- Earth turns towards east and makes complete rotation in 24hrs
- Turns 15 degrees in one hour and 0.25 degrees in one minute
- Prime meridian, all we need is a time marker
- 4 hours=60 degrees
- 18 min=4.5 degrees
- So ships is at 64.5 degrees W
  *question like this on test

Why was this method problematic?

- Clocks back then were pendulum driven which did not work at sea (spins)
- Find a better way to find longitude
- Watch maker was who invented a way (chronometer in 1761; watch that worked independently of pendulums)